

One-photon pseudoscalar feed-down in the v3 analysis

100k physics-model events per mode; frozen 1,091-chunk BDT proposal

FCC-ee $\Lambda_b \rightarrow \Lambda \gamma$ analysis

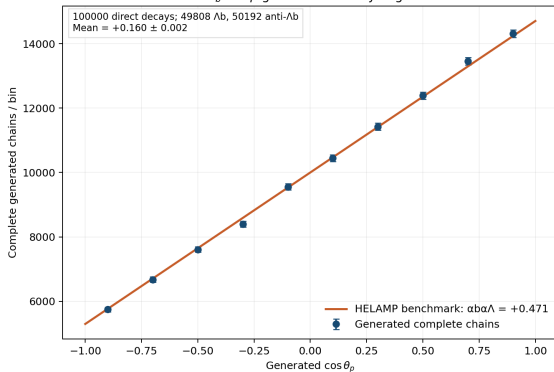
4 October 2026

Question and exact comparison

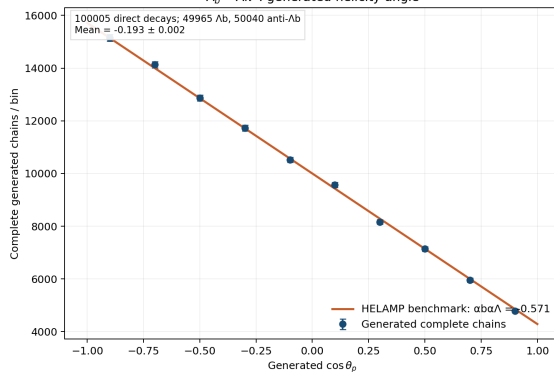
- Forced $\Lambda_b \rightarrow \Lambda \eta(\gamma\gamma)$ and $\Lambda_b \rightarrow \Lambda \pi^0(\gamma\gamma)$ HELAMP samples; both charge conjugates, 100,000 input events each.
- Same one-photon Stage 1 v3 builder, Λ mass ± 12.5 MeV, $E_{\Lambda_b} \geq 10.5$ GeV, same hemisphere, no meson veto.
- Validation-fixed BDT score ≥ 0.9538269639 ; physical estimates use candidate rows in a provisional 5.4–5.9 GeV reconstructed signal window.
- Truth ancestry labels direct partial and wrong combinations after reconstruction and selection.

Generated model checks

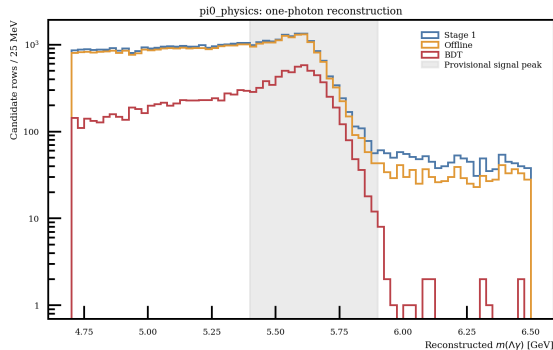
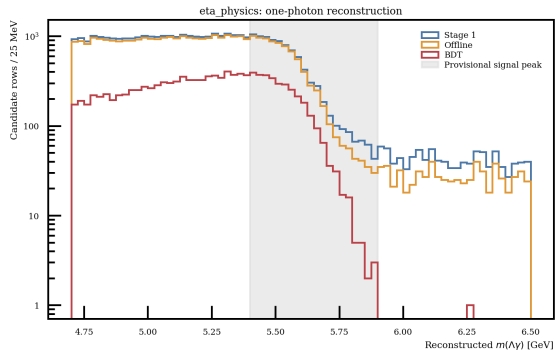
$\Lambda_b \rightarrow \Lambda \eta$: generated helicity angle



$\Lambda_b \rightarrow \Lambda \pi^0$: generated helicity angle

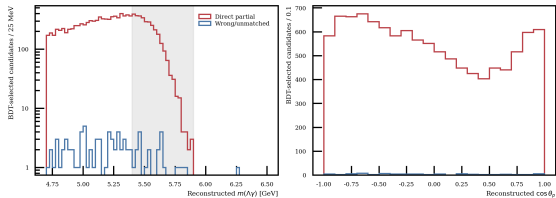


Stage-by-stage reconstructed mass

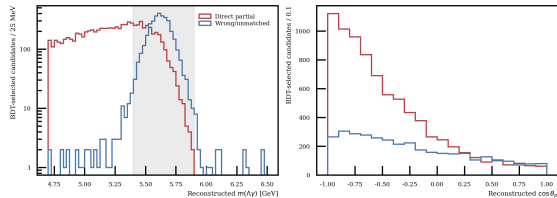


Surviving ancestry and angular shape

eta_physics: partial and wrong-combination components



pi0_physics: partial and wrong-combination components

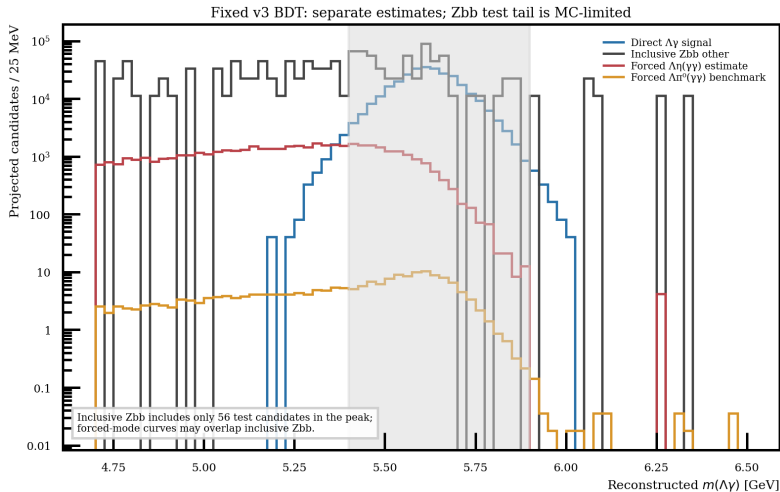


Candidate counts and denominators

Quantity	$\Lambda\eta$	$\Lambda\pi^0$
Generated direct decays	100,000	100,005
Stage 1 candidate-bearing events	41,752	45,104
Stage 1 candidate rows	44,717	48,298
Offline candidate rows	35,267	39,639
BDT candidate rows	11,125	11,512
Peak candidate rows	3,120	5,937
Peak direct partial / wrong rows	3,099 / 21	2,519 / 3,418

Offline retention from Stage 1: 78.87% / 82.07%. BDT retention from offline: 31.55% / 29.04%. These are forced-sample candidate rates, before branching normalization.

Physical scenarios, shown separately from inclusive $Z \rightarrow b\bar{b}$



Interpretation and PI decision

- With $N_Z = 6 \times 10^{12}$ and stated production and daughter fractions: $\Lambda\eta$ contributes about 13,136 peak candidates using the LHCb parent-BR central value; $\Lambda\pi^0$ contributes about 107 using a theory parent-BR benchmark.
- At fixed branching inputs, approximate candidate-count MC 95% intervals: 12,679–13,605 and 104–109. Parent-BR uncertainty dominates the η estimate; π^0 has no measured parent input here.
- Inclusive $Z \rightarrow b\bar{b}$ can already contain these modes. Audit overlap before adding forced-mode estimates to it. The inclusive peak shape has only 56 held-out test candidates.
- Decide whether to include explicit feed-down shapes and adopt a peak window and branching scenarios; the BDT score remains a proposal.